Diagonalization and the Minimal Polynomial

From the class book: Chapter 5, Sections 5.3–5.4, and Chapter 6, Section 6.1

1. Key definitions

Characteristic polynomial. $\chi_A(x) = \det(xI - A)$. Its roots are the eigenvalues. The *algebraic multiplicity* $\mathrm{AM}(\lambda)$ is the multiplicity of λ as a root; the *geometric multiplicity* $\mathrm{GM}(\lambda) = \dim \ker(A - \lambda I) = n - \operatorname{rank}(A - \lambda I)$. Always $1 \le \mathrm{GM} \le \mathrm{AM}$.

Minimal polynomial. $m_A(x)$ is the monic polynomial of least degree with $m_A(A) = 0$.

Diagonalizable. $A \in M_n(F)$ is diagonalizable over F if $P^{-1}AP$ is diagonal for some invertible $P \in M_n(F)$; equivalently F^n has a basis of eigenvectors of A.

2. Main results

Theorem (Cayley–Hamilton)

- $\chi_A(A) = 0$. For 2×2: $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$.
- Use it to reduce powers: every A^k is a combination of $I, A, \ldots, A^{n-1}$; and if $\det A \neq 0$, A^{-1} is a polynomial in A.

Theorem (Minimal polynomial)

- m_A divides every polynomial that kills A; in particular $m_A \mid \chi_A$.
- m_A and χ_A have the same roots; only multiplicities differ.
- $\deg m_A = \dim \operatorname{span}\{I, A, A^2, \ldots\} \le n$.
- Similar matrices have the same m_A (the converse fails). For block diagonal A, $m_A = \operatorname{lcm}(m_{A_i})$ and $\chi_A = \prod \chi_{A_i}$.
- The exponent of $(x - \lambda)$ in m_A is the size of the largest Jordan block for λ.

Theorem (Criteria for diagonalizability)

- A is diagonalizable over $F \iff \chi_A$ splits over F and GM = AM for every eigenvalue $\iff \sum_\lambda \dim E_\lambda = n$.
- $\iff m_A$ is a product of *distinct* linear factors over $F \iff$ some polynomial with distinct roots in F kills A.
- n distinct eigenvalues $\Rightarrow$ diagonalizable (the converse fails: $A = I$).

Matrix	Diagonalizable?
$A^2 = A$, $A^2 = I$, $A^k = I$ (over $\mathbb{C}$), $A^3 = A$	always, over $\mathbb{C}$
real symmetric, Hermitian, unitary, normal	always
nilpotent $N \neq 0$	never
real with non-real eigenvalues	not over $\mathbb{R}$; over $\mathbb{C}$ iff m_A has distinct roots
real 2×2: $(\operatorname{tr})^2 > 4 \det$ / $<$ / $=$	yes over $\mathbb{R}$ / no over $\mathbb{R}$ / iff scalar

3. Methods

Method: finding m_A.

- (1) Factor $\chi_A = \prod (x - \lambda_i)^{a_i}$.
- (2) Test $\prod (x - \lambda_i)^{b_i}$, $1 \leq b_i \leq a_i$, in increasing degree; the first that kills A is m_A.
- (3) For AM 2: $b = 1$ if GM $= 2$, $b = 2$ if GM $= 1$. For AM 3: $b = 1, 2, 3$ according as GM $= 3, 2, 1$.

Method: deciding diagonalizability.

- (1) Distinct eigenvalues in F: yes.
- (2) For each repeated λ, compute GM $= n - \operatorname{rank}(A - \lambda I)$; if GM $<$ AM anywhere, no.
- (3) Or check whether $\prod (A - \lambda_i I) = 0$ over the *distinct* eigenvalues.
- (4) Triangular matrices: look at the entries joining equal diagonal entries.

4. Exam shortcuts and common traps

- J_n (all ones): eigenvalues $n, 0, \ldots, 0$, $m = x(x - n)$; so $aI + bJ_n$ has $m = (x - a)(x - a - nb)$ for $b \neq 0$.
- $A^2 = 0$, $A \neq 0$, $n = 3$: rank $A = 1$, since $\operatorname{im} A \subseteq \ker A$ forces $2 \operatorname{rank} A \leq 3$.
- $A^3 = A \Rightarrow$ eigenvalues in $\{0, \pm 1\}$, A diagonalizable and A^2 idempotent.
- **Traps:** same χ does not mean similar; same m does not mean similar; diagonalizable over $\mathbb{C}$ is not over $\mathbb{R}$; a matrix with repeated eigenvalues can still be diagonalizable.

5. Solved examples

Example 1. Find m_A for $A = \operatorname{diag}\left(\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, 3 \right)$.

Solution

- The blocks have minimal polynomials $(x - 3)^2$, $(x - 3)(x + 1)$, $x - 3$.
- $m_A = \operatorname{lcm} = (x - 3)^2 (x + 1)$, while $\chi_A = (x - 3)^4 (x + 1)$.

Example 2. When is $M = \begin{pmatrix} 2 & a & b \\ 0 & 3 & c \\ 0 & 0 & 2 \end{pmatrix}$ diagonalizable?

Solution

- Eigenvalues $2, 2, 3$. $\text{GM}(2) = 3 - \operatorname{rank} \begin{pmatrix} 0 & a & b \\ 0 & 1 & c \\ 0 & 0 & 0 \end{pmatrix}$.
- This rank is 1 iff (a, b) is a multiple of $(1, c)$, i.e. $b = ac$. So M is diagonalizable iff $b = ac$.

Example 3. Let $A \in M_2(\mathbb{R})$ with $\operatorname{tr} A = 3$, $\det A = 2$. Express A^3 and A^{-1} in terms of A and I.

Solution

- Cayley–Hamilton: $A^2 = 3A - 2I$, so $A^3 = 3A^2 - 2A = 7A - 6I$.
- $A(3I - A) = 2I$ gives $A^{-1} = \frac{1}{2}(3I - A)$.

Example 4. Find the minimal polynomial of $A = J + 2I$, where J is the 6×6 all-ones matrix.

Solution

- $J^2 = 6J$, so $m_J = x(x - 6)$, and A has eigenvalues 8 (once) and 2 (five times).
- A is symmetric, hence diagonalizable: $m_A = (x - 2)(x - 8)$.

Example 5. Which of $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ are diagonalizable over $\mathbb{R}$?

Solution

- The first has eigenvalues $\pm i$: not over $\mathbb{R}$ (but yes over $\mathbb{C}$).
- The second is a non-scalar matrix with $(\operatorname{tr})^2 = 4 \det$: not diagonalizable.
- The third has $(\operatorname{tr})^2 = 25 > 4 \det = -8$: two distinct real eigenvalues, so yes.

Example 6. Let $A \in M_n(\mathbb{C})$ with $A^3 = A$. What can you say?

Solution

- $x^3 - x = x(x - 1)(x + 1)$ has distinct roots and kills A: A is diagonalizable with eigenvalues in $\{0, 1, -1\}$.
- $(A^2)^2 = A \cdot A^3 = A^2$, so A^2 is idempotent. A need not be invertible ($A = 0$).